

EDMONTON INTL, AB, Canada

WMO: 711230

Lat: 53.3067N

Lon: 113.6058W

Elev: 723

StdP: 92.93

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.3	(c) -29.0	(d) -36.5	(e) 0.1	(f) -32.2	(g) -33.1	(h) 0.2	(i) -28.9	(j) 11.3	(k) -3.7	(l) 10.0	(m) -7.7	(n) 2.1	(o) 190	(p) 0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.7	(c) 27.8	(d) 18.0	(e) 26.1	(f) 17.3	(g) 24.5	(h) 16.4	(i) 19.9	(j) 25.7	(k) 18.6	(l) 24.3	(m) 17.5	(n) 22.8	(o) 3.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	13.9	23.0	16.4	12.8	21.5	15.3	11.9	20.3	60.3	25.8	55.9	24.2	52.2	22.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.3	(b) 8.9	(c) 7.8	(e) -37.8	(f) 31.3	(g) 4.0	(h) 2.0	(i) -40.7	(j) 32.7	(k) -43.1	(l) 33.8	(m) -45.4	(n) 35.0	(o) -48.3	(p) 36.4		
			DB													
			WB													
				-37.9	22.3	4.0	1.6	-40.8	23.4	-43.2	24.3	-45.4	25.2	-48.3	26.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-12.1	-10.6	-5.3	3.4	10.1	14.1	16.3	15.1	10.4	3.6	-5.1	-11.1
	DBStd	11.98	8.91	7.75	7.89	5.30	4.49	3.01	2.83	3.14	4.25	4.61	7.00	8.06
	HDD10.0	3354	684	577	476	203	54	4	1	2	44	202	453	654
	HDD18.3	5814	943	810	733	448	256	132	75	108	238	456	703	912
	CDD10.0	605	0	0	0	5	57	126	196	159	58	4	0	0
	CDD18.3	24	0	0	0	0	1	3	12	7	1	0	0	0
	CDH23.3	609	0	0	0	2	58	83	220	174	68	2	0	0
	CDH26.7	109	0	0	0	0	9	12	42	32	14	0	0	0
(14) Wind	WSAvg	3.5	3.3	3.3	3.7	4.1	4.2	3.7	3.1	2.9	3.3	3.7	3.4	3.4
(15) Precipitation	PrecAvg	493	25	16	17	26	48	84	98	70	47	21	19	19
	PrecMax	651	74	40	46	61	136	170	209	199	144	69	60	46
	PrecMin	357	4	1	3	3	10	13	19	8	3	0	1	2
	PrecStd	76	19	10	10	15	25	43	38	40	30	15	13	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	6.3	12.4	22.7	28.0	28.2	30.3	29.8	28.6	21.9	12.6	7.3
		MCWB	2.7	2.5	6.7	11.4	14.3	17.5	20.2	19.3	16.9	12.3	7.0	3.1
	2%	DB	4.0	3.7	8.4	18.9	24.8	25.6	27.6	27.2	25.2	17.3	8.2	4.4
		MCWB	1.7	1.0	3.9	9.3	13.1	16.0	19.4	18.4	15.3	9.9	4.2	1.4
	5%	DB	2.1	2.0	5.7	15.6	22.4	23.7	25.8	25.2	22.4	14.1	5.5	2.4
		MCWB	0.3	-0.2	2.5	7.7	12.2	15.2	18.7	17.6	14.2	8.0	2.5	0.1
	10%	DB	-0.1	0.2	3.9	12.8	19.9	21.7	24.0	23.3	19.5	11.6	3.4	0.1
		MCWB	-1.7	-1.5	1.3	6.1	10.9	14.2	17.4	16.8	12.7	6.7	1.2	-1.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.9	7.0	11.7	15.4	19.2	22.2	20.8	17.6	12.9	7.4	3.6
		MCDB	5.4	5.8	11.6	21.2	24.8	24.8	27.0	26.8	25.9	20.9	12.0	6.8
	2%	WB	1.8	1.3	4.2	9.8	14.0	17.5	20.5	19.4	15.8	10.4	4.7	1.7
		MCDB	3.7	3.5	7.7	18.1	22.5	23.3	26.1	25.4	23.8	16.1	7.6	4.0
	5%	WB	0.4	-0.1	2.7	8.2	12.8	16.2	19.2	18.3	14.6	8.6	2.8	0.2
		MCDB	2.1	1.8	5.5	14.8	21.0	21.8	24.7	24.1	21.8	13.2	5.1	2.3
	10%	WB	-1.7	-1.6	1.5	6.6	11.5	15.1	18.0	17.2	13.1	7.1	1.4	-1.6
		MCDB	-0.2	0.2	3.6	11.9	18.8	20.2	23.0	22.5	18.7	11.0	3.2	0.1
(35) Mean Daily Temperature Range	MDBR		10.2	11.3	10.5	12.2	13.8	12.1	12.7	13.6	14.0	12.2	9.7	10.1
	5% DB	MCDBR	11.3	12.5	12.1	18.4	18.2	16.2	16.2	17.4	19.7	17.2	12.1	11.1
		MCWBR	9.5	10.4	8.6	10.4	9.1	8.6	9.5	9.9	10.8	10.7	8.7	8.9
	5% WB	MCDBR	11.0	11.9	11.5	17.2	16.4	13.7	15.0	15.9	18.4	15.8	11.5	11.0
		MCWBR	9.3	10.0	8.3	10.1	8.5	8.1	9.3	9.7	10.2	10.1	8.6	9.1
(40) Clear-Sky Solar Irradiance	taub		0.257	0.265	0.308	0.326	0.342	0.350	0.353	0.361	0.307	0.282	0.254	0.241
	taud		2.315	2.373	2.337	2.390	2.393	2.388	2.421	2.354	2.497	2.545	2.461	2.318
	Ebn at Noon		739	859	881	908	907	900	889	860	878	835	755	690
	Edh at Noon		57	74	97	105	110	112	107	108	82	63	49	47
(44) All-Sky Solar Radiation	RadAvg		1.00	2.04	3.56	4.68	5.72	6.03	6.20	5.10	3.53	2.05	1.00	0.73
	RadStd		0.10	0.16	0.27	0.45	0.40	0.39	0.31	0.35	0.37	0.24	0.11	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.51	N/A	N/A	N/A
(47) Regional (73 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page